SynthMicro — Whole Blood Smear XAI Report

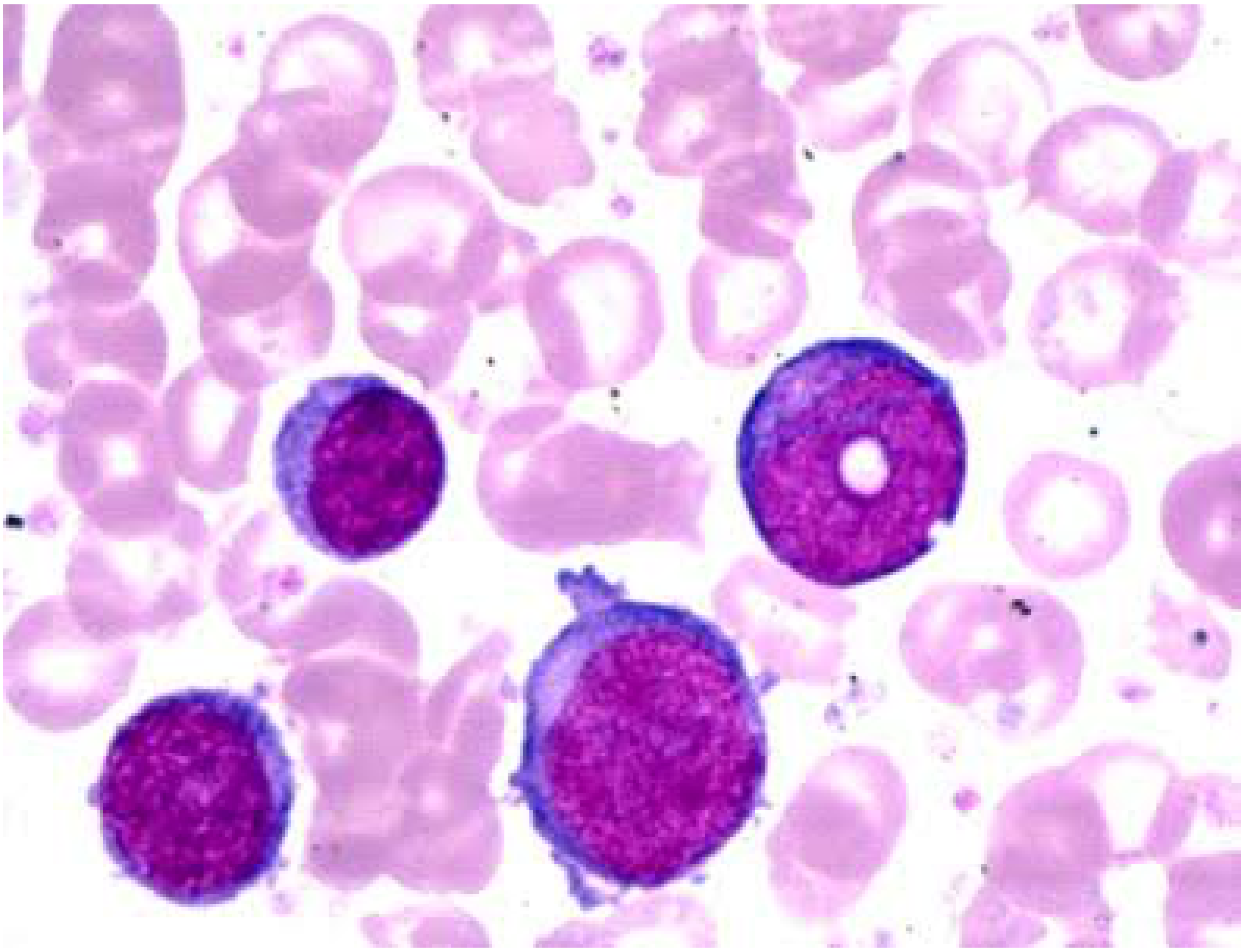
Analysis ID: ce37e6353b4a456ca966c1e039e77b34
Generated: 2026-09-08 12:08:34

Pipeline: Cellpose segmentation → 384×384 cell extraction → Cellpose → ResNet50 → Grad-CAM → optional SHAP → optional VLM → PDF report.

Measurement	Result
Cellpose objects	4
Nucleated-cell candidates	4
Myeloblast-like candidates	4
Myeloblast-like proportion	100.00%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

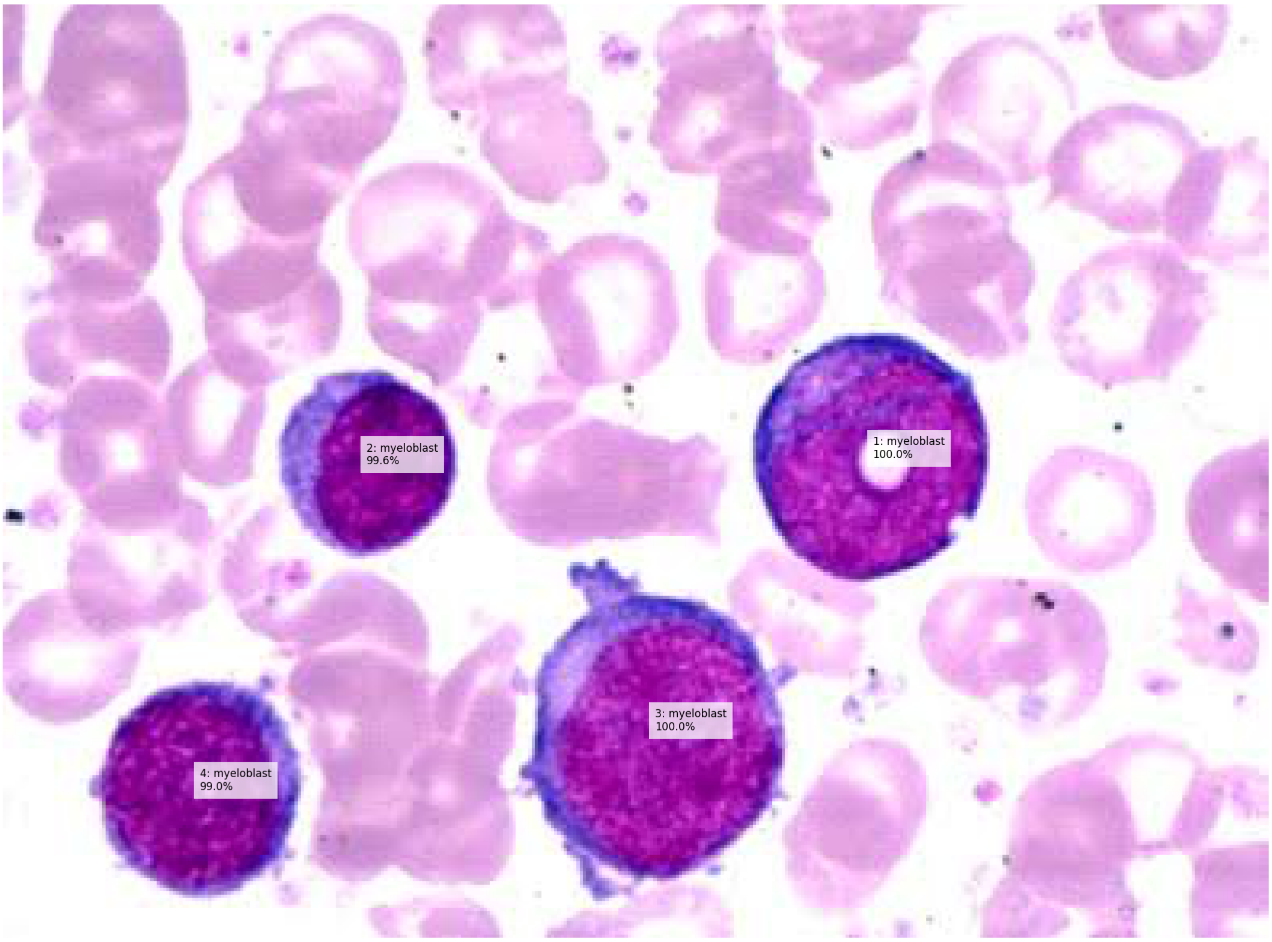
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



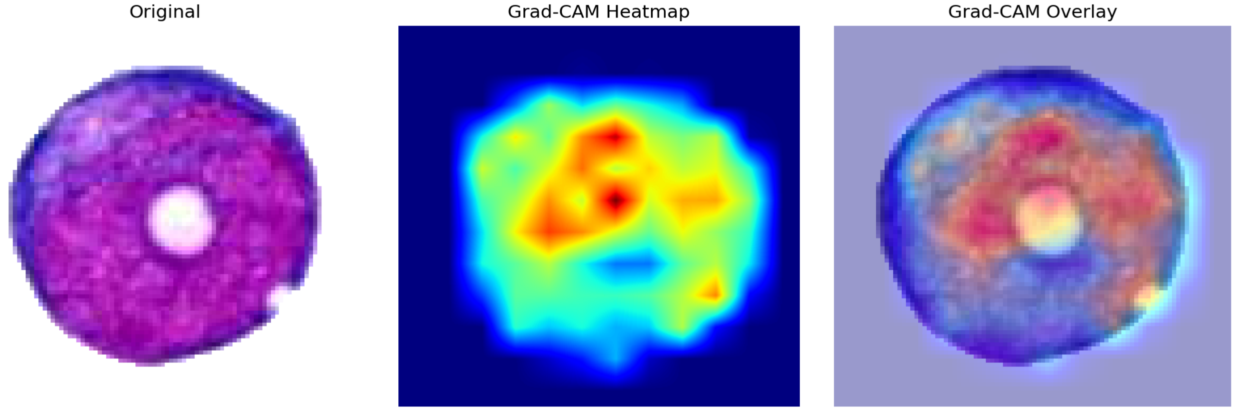
Cell-Level Results

ID	Area	Purple	Class	Confidence
1	4324	120.9	myeloblast	100.0%
2	2470	107.6	myeloblast	99.6%
3	5848	117.2	myeloblast	100.0%
4	3325	114.2	myeloblast	99.0%

Grad-CAM Explanations

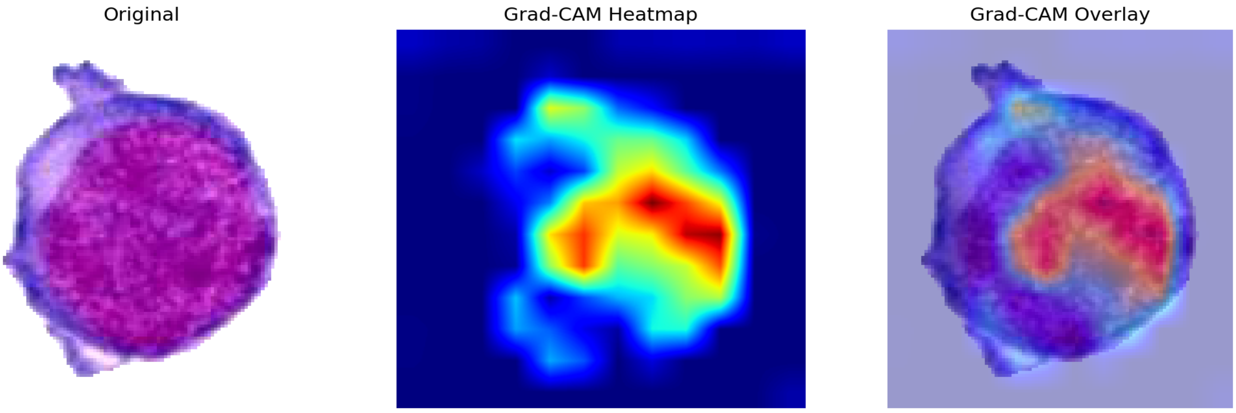
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 1 — myeloblast (100.0%)



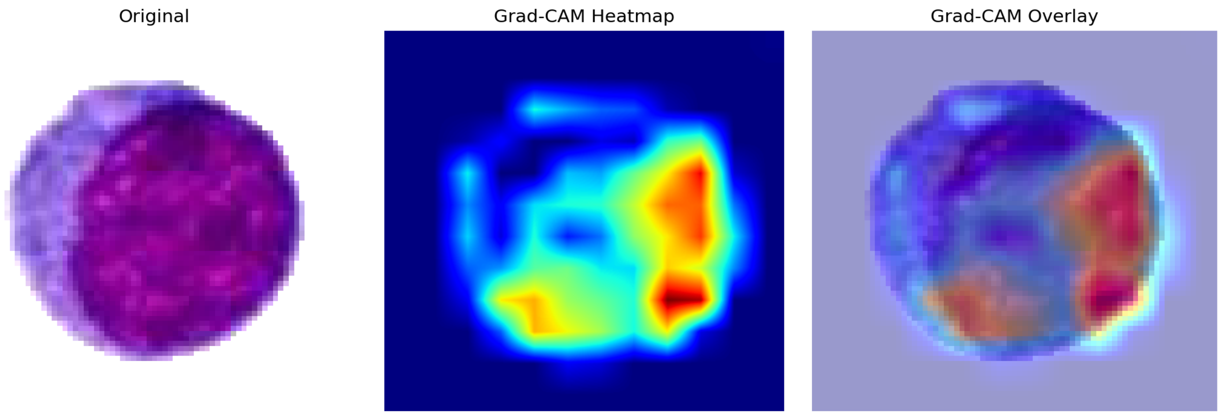
The ResNet50 model classified this segmented candidate as a myeloblast with 100.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 3 — myeloblast (100.0%)



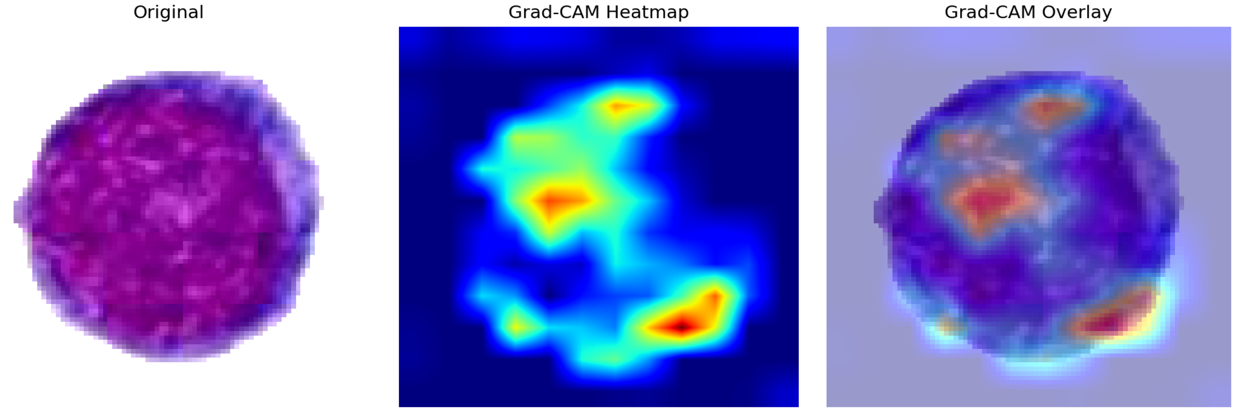
The ResNet50 model classified this segmented candidate as a myeloblast with 100.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 2 — myeloblast (99.6%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.6% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 4 — myeloblast (99.0%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

- The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.
- RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.
- Cellpose segmentation quality affects every downstream result.
- Stain, microscope, illumination and acquisition differences can cause domain shift.
- Confidence is not a calibrated probability of disease.
- A myeloblast-like prediction is not equivalent to an AML diagnosis.
- Clinical assessment requires appropriate hematopathology and laboratory testing.